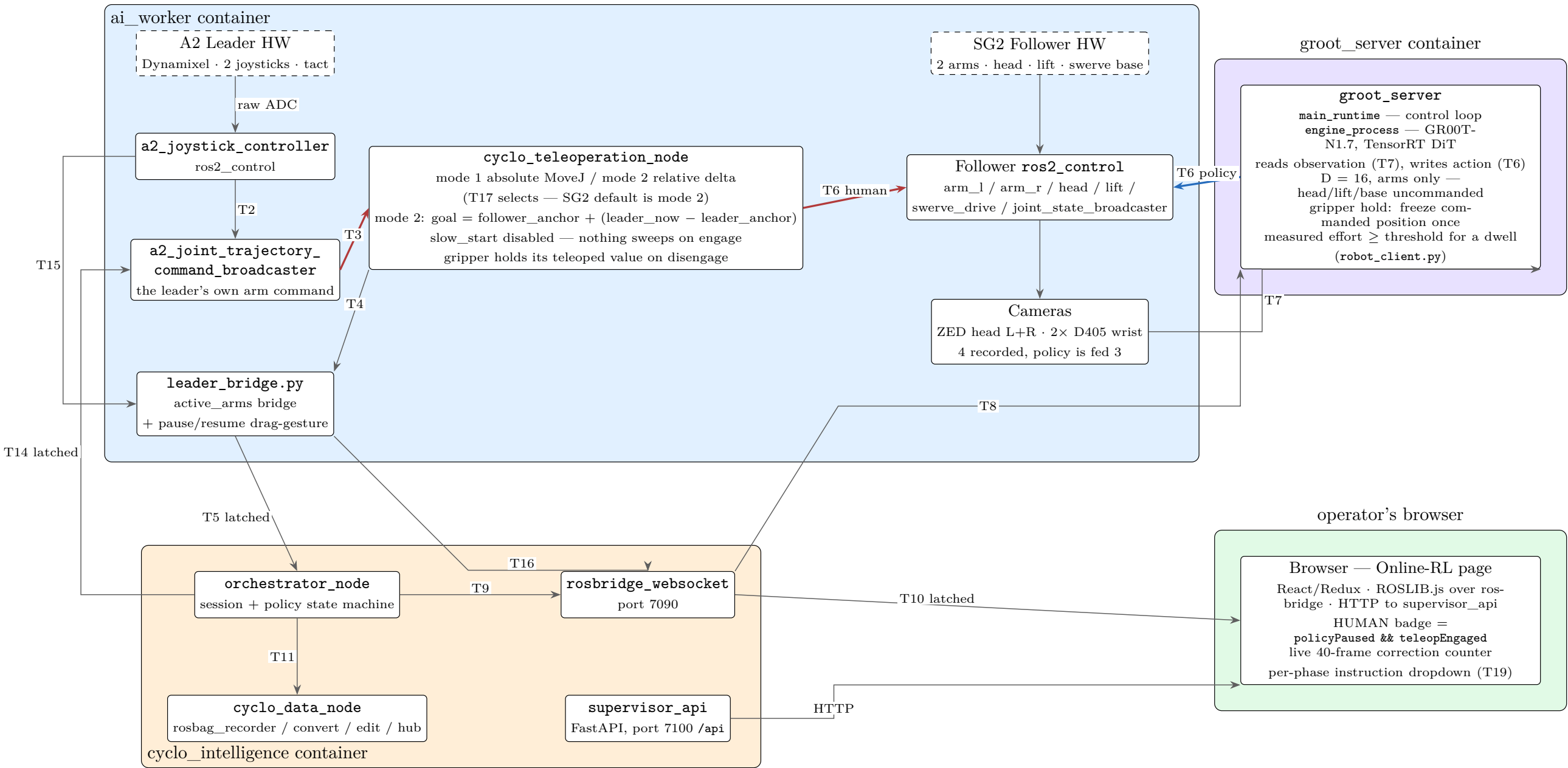


# HG-Dagger / Online-RL — Runtime Control Path

FFW SG2 follower · A2 leader · GR00T-N1.7 policy

— human commands the arms — policy commands the arms — status / feedback / control plane. Edge labels are IDs defined in the legend on p.3.



**The operator's takeover loop**

- joystick drag-down  $\geq 1s$  → T16 → INFERENCING → PAUSED
- tact press → T2 COMMAND\_TOGGLE → that arm engages
- tact press again → arm disengages, gripper holds its teleoped value
- joystick drag-down  $\geq 1s$  → T16 → PAUSED → INFERENCING

The gaps 1→2 and 3→4 are PAUSED with no arm engaged: labelled -1, not human.

T16 is one topic with two meanings, resolved by the phase it arrives in.

T6 is ONE topic, written by BOTH sides  
`/leader/joint_trajectory_command_broadcaster_{left,right}/joint_trajectory`

Teleop node and policy publish to the same place, so *action* is valid in either mode — no stale-leader pose. T14 stops them contending.

T18 Home goes straight from the browser to *ai\_worker* over rosbridge — the only UI action that skips the orchestrator.

# Recording and Conversion — what survives to training

Every recorded topic, and where the online-RL labels are lost

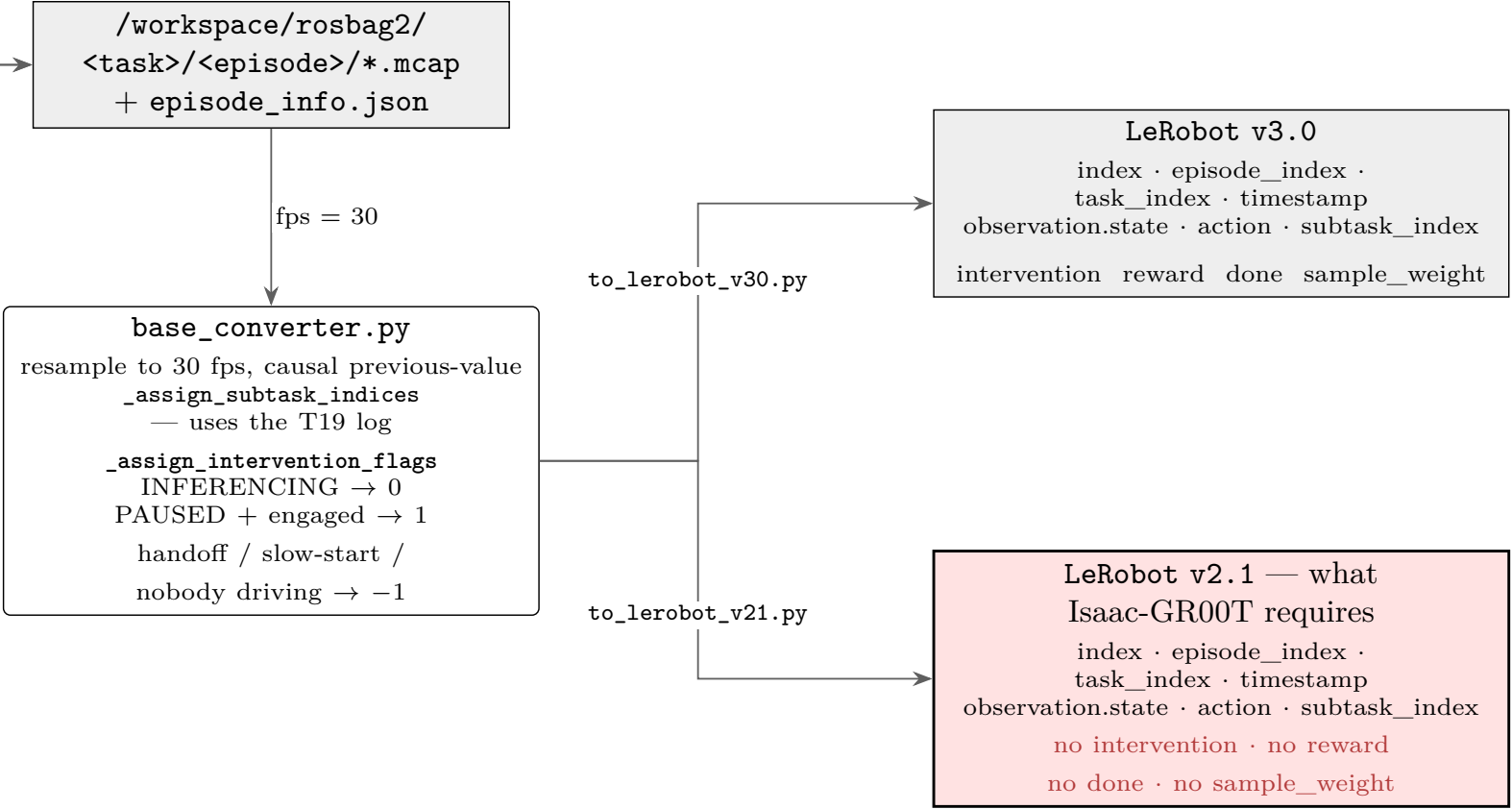
recorded topics — `ffw_sg2_rev1_config.yaml`

observation.images — 4 cameras  
cam\_left\_head /zed/zed\_node/left/image\_rect\_color/compressed  
cam\_left\_wrist /camera\_left/.../image\_rect\_raw/compressed  
rot 270  
cam\_right\_wrist /camera\_right/.../image\_rect\_raw/compressed  
rot 270  
cam\_right\_head  
/zed/zed\_node/right/image\_rect\_color/compressed

observation.state — 22 dims  
upper\_body /joint\_states JointState — 19 joints:  
arm\_l\_joint1..7, gripper\_l\_joint1,  
arm\_r\_joint1..7, gripper\_r\_joint1,  
head\_joint1..2, lift\_joint  
mobile /odom Odometry — linear\_x, linear\_y, angular\_z

action — 20 dims. Each topic is BOTH the policy’s command  
target and the record target.  
arm\_left  
/leader/joint\_trajectory\_command\_broadcaster\_left/...  
arm\_right  
/leader/joint\_trajectory\_command\_broadcaster\_right/...  
head /leader/joystick\_controller\_left/joint\_trajectory  
lift /leader/joystick\_controller\_right/joint\_trajectory  
mobile /cmd\_vel Twist

recording.extra\_topics — in the bag, not training data  
/task/inference\_status ← the intervention source  
/tf  
/zed/zed\_node/{left,right}/camera\_info  
/camera\_{left,right}/.../camera\_info  
/leader/teleoperation/control\_status deliberately absent —  
robotis\_interfaces is not installed where the recorder runs.



This is where HG-Dagger dies. `_assign_intervention_flags` computes a correct per-frame label, and `to_learnerobot_v30.py` writes it out (lines 1484–1489). `to_learnerobot_v21.py` never does — `grep -c intervention` on it returns 0. Isaac requires v2.1, so every GR00T export silently loses the human/policy label and degrades to plain behaviour cloning. Fix: *IMPROVEMENTS.md* §3.

Topic / Service Legend

Verified against cyclo\_intelligence\_backup@main and aiw\_backup@feature-a2, 2026-08-30.

| ID  | Kind               | Name                                                                           | Type                                            | Meaning / gotcha                                                                                                                                                                                       |
|-----|--------------------|--------------------------------------------------------------------------------|-------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| T1  | PUB/SUB            | /leader/joystick_controller/tact_trigger                                       | std_msgs/String                                 | Tap emits "left"/"right". Record trigger disabled — one press also engages that arm, which during INFERENCING would put the leader onto T6 against the policy.                                         |
| T2  | PUB/SUB            | .../joystick_command                                                           | robotis_interfaces/<br>TeleoperationCommand     | ENABLE / DISABLE / TOGGLE, per arm. The tact sends TOGGLE — engage and stop are the same press.                                                                                                        |
| T3  | PUB/SUB            | .../raw_joint_trajectory                                                       | trajectory_msgs/<br>JointTrajectory             | The leader’s own arm command, before mode handling.                                                                                                                                                    |
| T4  | PUB/SUB            | /leader/teleoperation/control_status                                           | robotis_interfaces/<br>ControlModeStatus        | <b>active_arms</b> + mode. Not recorded: <b>robotis_interfaces</b> is absent in the recorder, and adding it killed the launch group. Use T5.                                                           |
| T5  | PUB/SUB<br>latched | /leader/teleoperation/active_arms_str                                          | std_msgs/String                                 | Same information as T4 as a plain string, from <b>leader_bridge.py</b> . The UI badge reads it; the bag should too.                                                                                    |
| T6  | PUB/SUB            | /leader/joint_trajectory_command_<br>broadcaster_{left,right}/joint_trajectory | trajectory_msgs/<br>JointTrajectory             | Written by BOTH the teleop node and the policy. That is why <b>action</b> is valid in either mode. T14 prevents contention, not separate topics.                                                       |
| T7  | PUB/SUB            | /joint_states, /odom, 4 camera topics                                          | JointState, Odometry,<br>CompressedImage        | Follower feedback and vision. 4 cameras recorded, 3 declared to the policy.                                                                                                                            |
| T8  | PUB/SUB            | /<backend>/engine_command                                                      | std_msgs/String                                 | Orchestrator → <b>groot_server</b> : load / unload / run.                                                                                                                                              |
| T9  | SERVICE            | /task/command                                                                  | SendCommand.srv                                 | Start / Pause / Resume / Clear policy; Start / Stop / Cancel recording.                                                                                                                                |
| T10 | PUB/SUB<br>latched | /task/inference_status                                                         | interfaces/InferenceStatus                      | <b>inference_phase</b> : READY 0, LOADING 1, INFERENCING 2, PAUSED 3. Transitions only — the converter forward-fills. Sole source of the intervention label.                                           |
| T11 | SERVICE            | /data/recording                                                                | RecordingCommand.srv                            | Recording lifecycle, orchestrator → <b>cyclo_data_node</b> .                                                                                                                                           |
| T13 | SERVICE            | /data/convert, /data/convert/status                                            | StartConversion.srv,<br>GetConversionStatus.srv | LeRobot conversion. Both <b>convert_v21</b> and <b>convert_v30</b> false means "run both".                                                                                                             |
| T14 | PUB/SUB<br>latched | /task/policy_active                                                            | std_msgs/Bool                                   | Mirrors T10’s INFERENCING into <b>ai_worker</b> . Gates the leader chain off T6 so the two never publish together.                                                                                     |
| T15 | PUB/SUB            | .../sensorxel_joy_values                                                       | std_msgs/Float64MultiArray                      | Raw joystick axes. The gesture detector watches these.                                                                                                                                                 |
| T16 | PUB/SUB            | /leader/teleoperation/toggle_pause_request                                     | std_msgs/Empty                                  | Fires once on a ≥1s drag-hold. One topic, two meanings: pause if INFERENCING, resume if PAUSED. Lets the operator work without the browser.                                                            |
| T17 | PUB/SUB            | .../control_command                                                            | robotis_interfaces/<br>ControlModeCommand       | Mode 1 absolute MoveJ, mode 2 relative delta. SG2 default is mode 2.                                                                                                                                   |
| T18 | PUB/SUB<br>latched | /task/home_in_progress                                                         | std_msgs/Bool                                   | One-shot Home trajectory, published straight from the browser — the only UI action that skips the orchestrator.                                                                                        |
| T19 | SERVICE            | /task/command (UPDATE_INSTRUCTION)                                             | SendCommand.srv                                 | Appends to <b>instruction_change_log</b> , which splits the episode into correctly-timed sub-segments. Without it a continuous take gets one segment and every frame carries the last instruction set. |

Derived dataset columns

- **observation.state** (22) and **action** (20) — they differ: state carries **head\_joint1/2**, action does not. Slice order follows YAML insertion order. The policy declares only the first 16 (both arms); the remainder are recorded but not trained on.
- **task\_index** / **subtask\_index** — per frame, from **episode\_info.json**’s **segments**, split by **instruction\_change\_log** (T19). The episode’s top-level **task\_instruction** is a separate stable field; it is not the live per-phase instruction, which is overwritten on every T19 click.
- **intervention** — 1 = human drove, 0 = policy drove, −1 = excluded (handoff, nobody driving, phase never observed — note the handoff rule assumes a mode-1 sweep, see **IMPROVEMENTS.md** Defect 2). This is the inverse of **task\_is\_policy**, the name some downstream tooling expects; getting the polarity wrong raises nothing and trains on exactly the wrong frames. v3.0 only.
- **reward**, **done**, **sample\_weight** — emitted alongside **intervention**. v3.0 only, so HIL-SERL-style training is also impossible from a v2.1 export.